The rebalancing engine: one pass per trading day

Signal from yesterday, trade at today's price, pay the cost, record, repeat

1. Observe
BAR t ARRIVES

- Execution prices $S_i(t)$, known at the close
- Signal from closes through $t-1$: EMAs $\bar{S}^{\text{short}}_{t-1}$, $\bar{S}^{\text{long}}_{t-1}$ give ξ_t
- SIM parameters α_i, β_i and g'_M, frozen today (L7b updates them)

2. Orient
PREFERENCES

Preference weight of firm i:

$$\gamma_i(t) = \tanh\left(\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t} g'_M\right)$$

- sign: $\gamma_i > 0$ is preferred ($\mathcal{A}^+$), otherwise non-preferred ($\mathcal{A}^-$)
- magnitude: share of the budget

3. Decide
TARGETS AND RULES

Cobb–Douglas targets on $\mathcal{A}^+$:

$$n_i^\star = \frac{\gamma_i}{\sum_{j\in\mathcal{A}^+}\gamma_j}\,\frac{W_{\text{adj}}}{S_i(t)}$$

- schedule: trade only if $t \in \mathcal{R}$
- turnover cap $\tau_{\max}$ scales the move
- drawdown limit $d_{\max}$: go to cash

4. Act
TRADE AND RECORD

- Trade Δn_i at $S_i(t)$
- Pay $C_t = c\sum_i |\Delta n_i|\, S_i(t)$ before the targets are sized
- $W^+_{\mathcal{P}}(t) = W^-_{\mathcal{P}}(t) - C_t$; update the cash
- Record positions, weights, cost

next bar $t+1$: cash accrues $e^{g_f \Delta t}$, mark to market at $S_i(t+1)$, check the drawdown breaker

A static allocation (L5b, L6b) is one pass on the decision date, then hold.